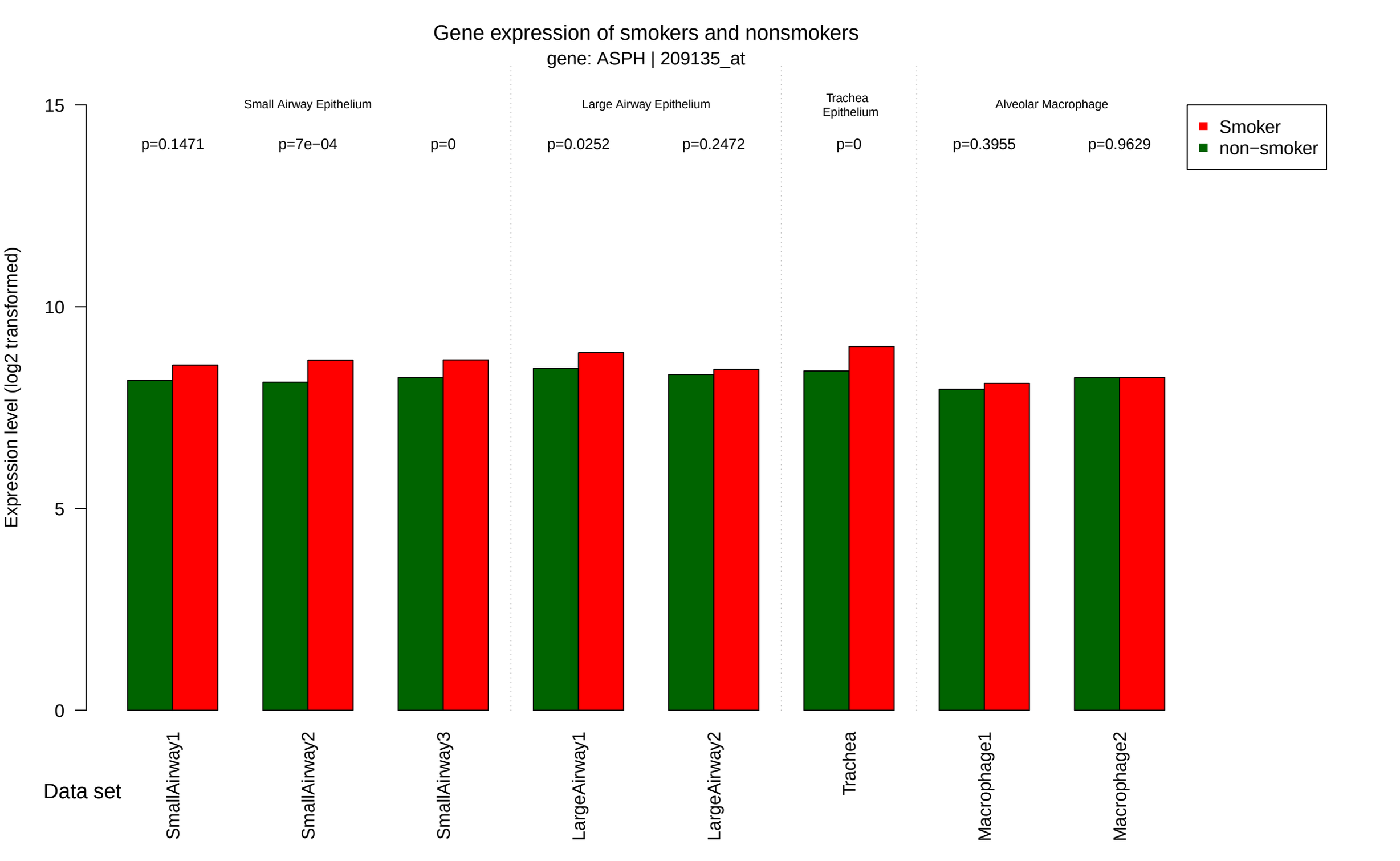
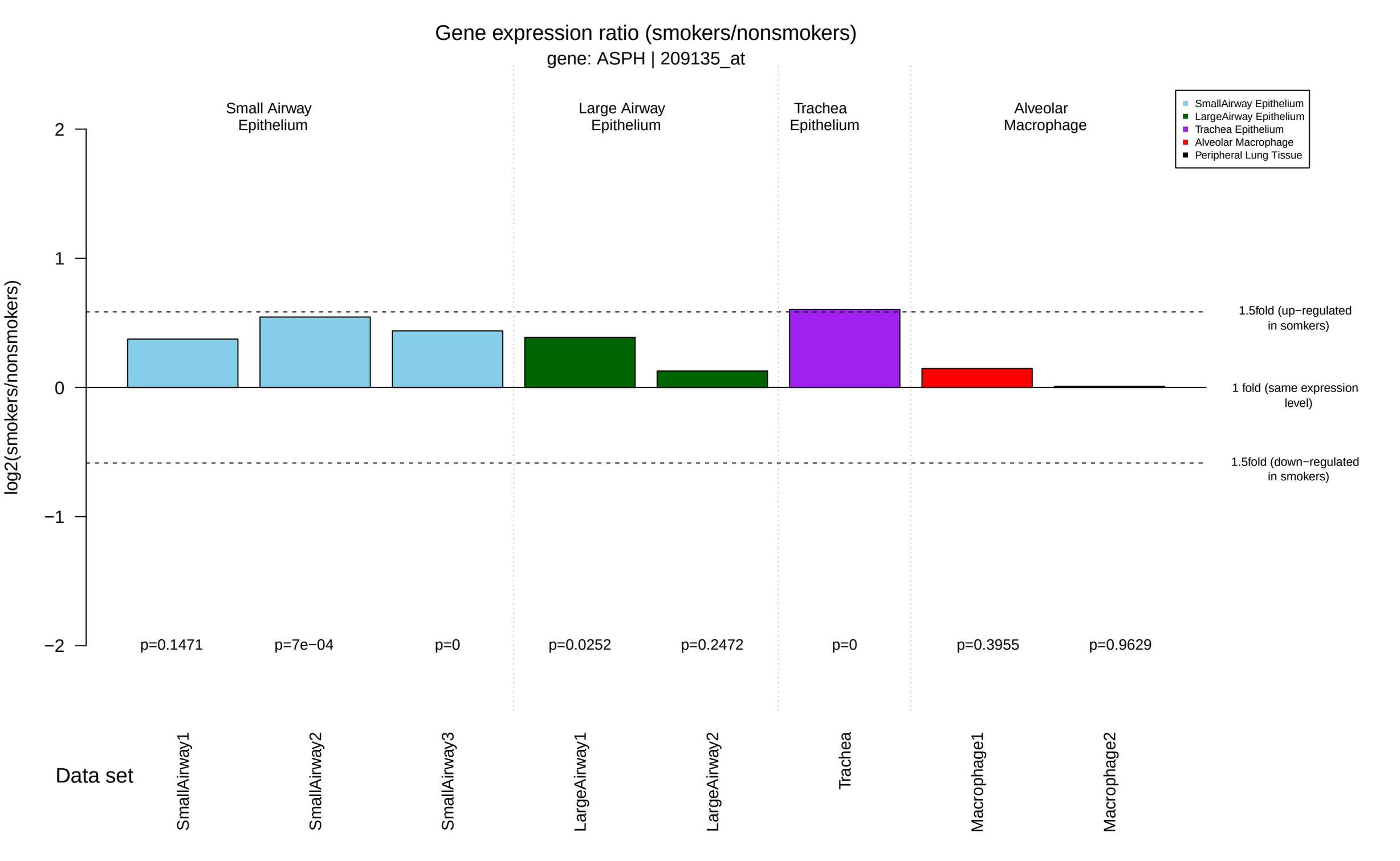
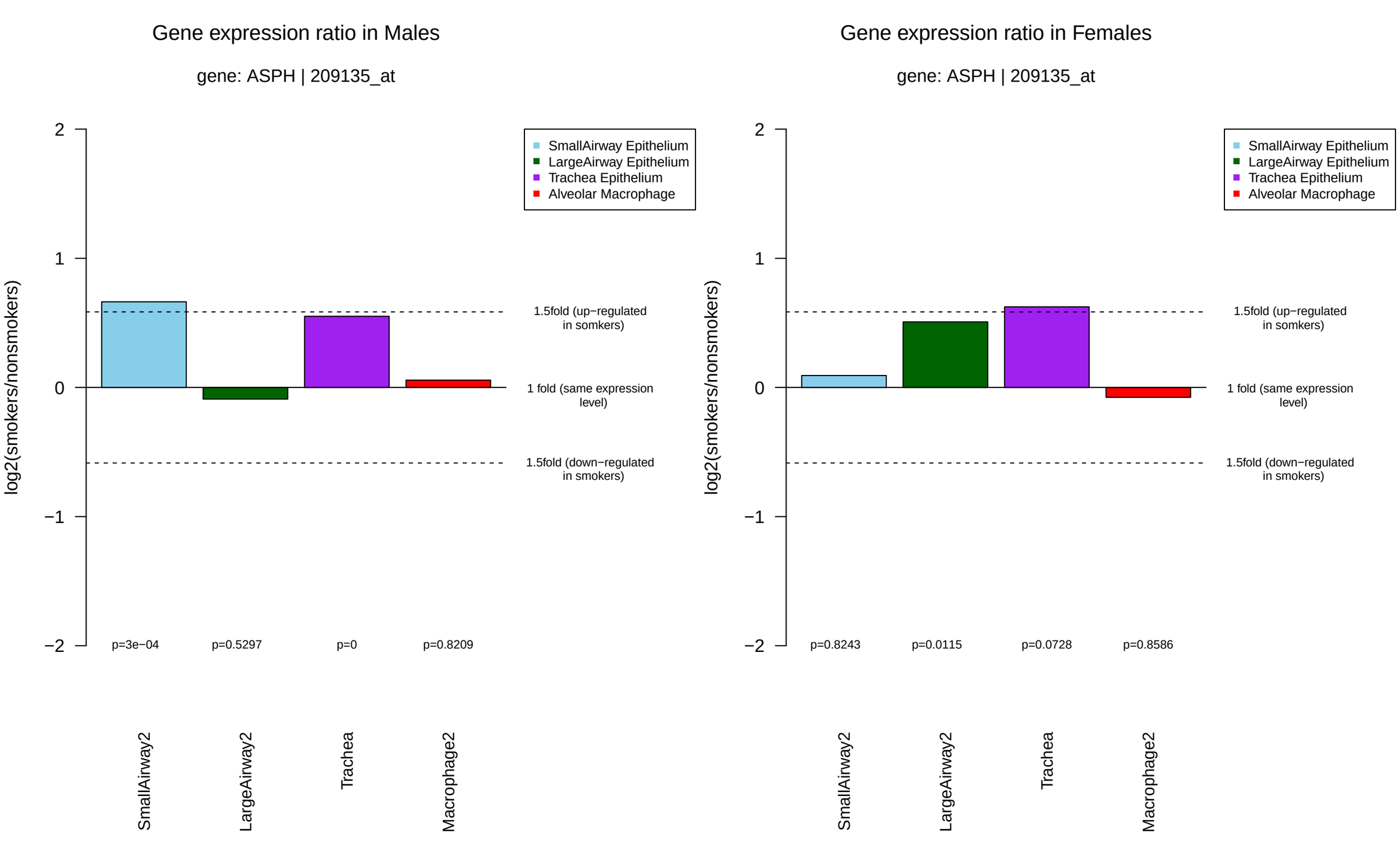
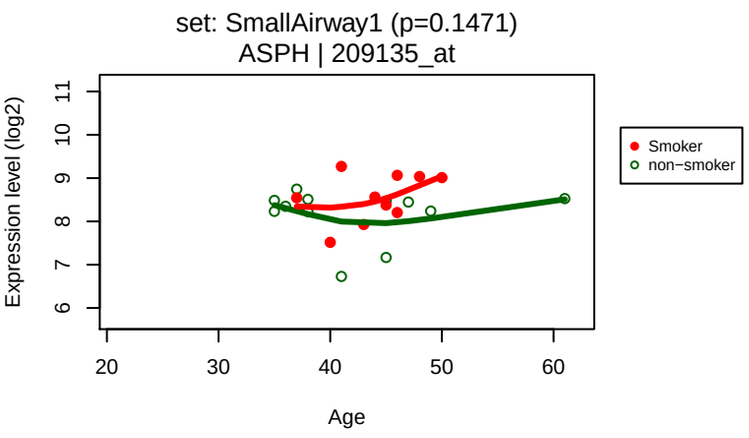
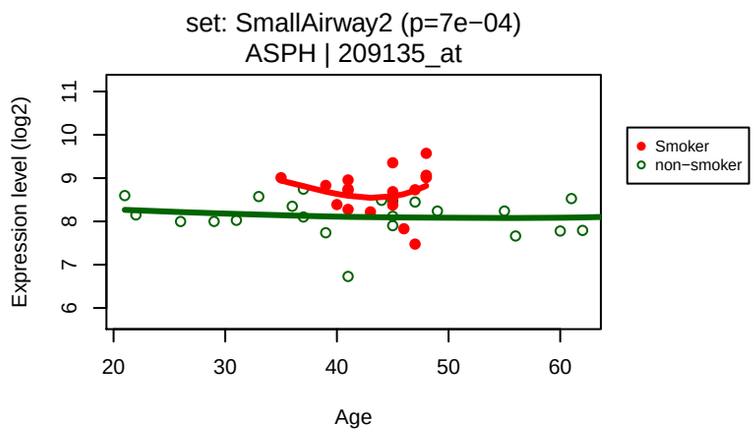
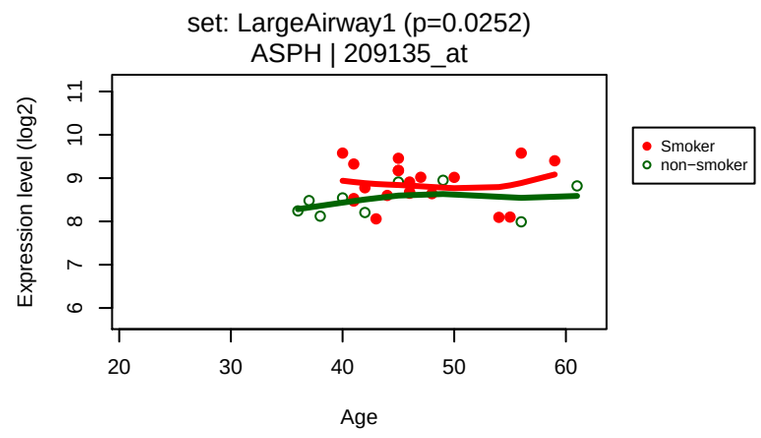

Gene expression of smokers and nonsmokers
gene: ASPH | 209135_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1471
p=7e−04
p=0
p=0.0252
p=0.2472
p=0
p=0.3955
p=0.9629
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

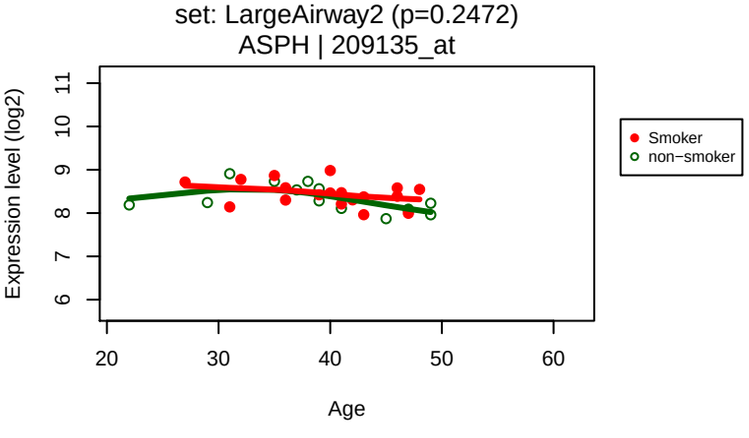

Gene expression ratio (smokers/nonsmokers)
gene: ASPH | 209135_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1471
p=7e−04
p=0
p=0.0252
p=0.2472
p=0
p=0.3955
p=0.9629
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

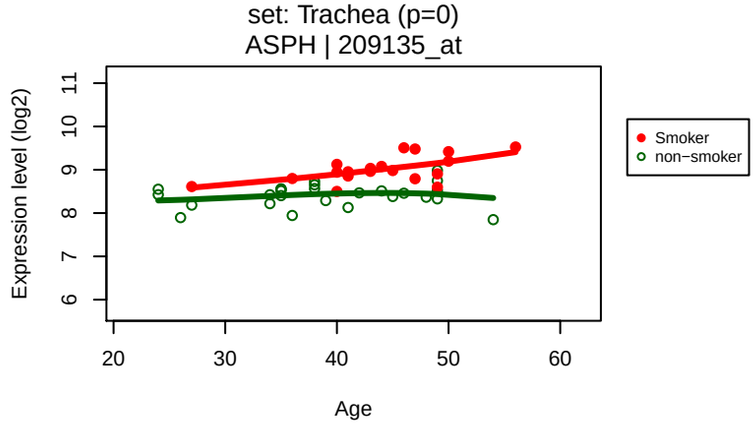

Gene expression ratio in Males
gene: ASPH | 209135_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=3e−04
p=0.5297
p=0
p=0.8209
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASPH | 209135_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8243
p=0.0115
p=0.0728
p=0.8586
SmallAirway2
LargeAirway2
Trachea
Macrophage2

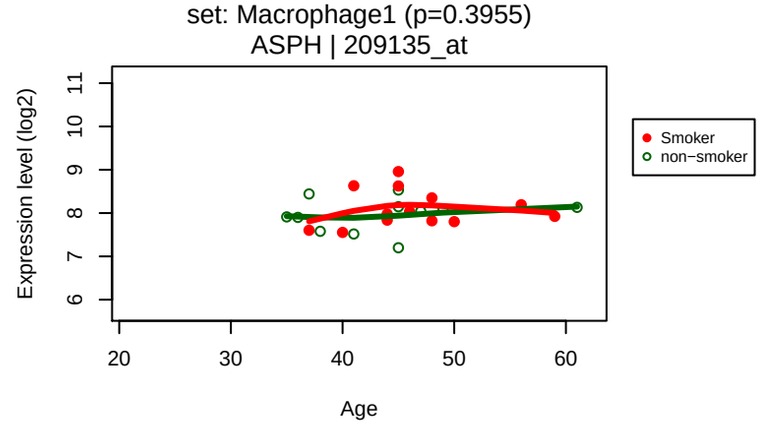

set: SmallAirway1 (p=0.1471)
ASPH | 209135_at
Expression level (log2)
Age
Smoker
non−smoker

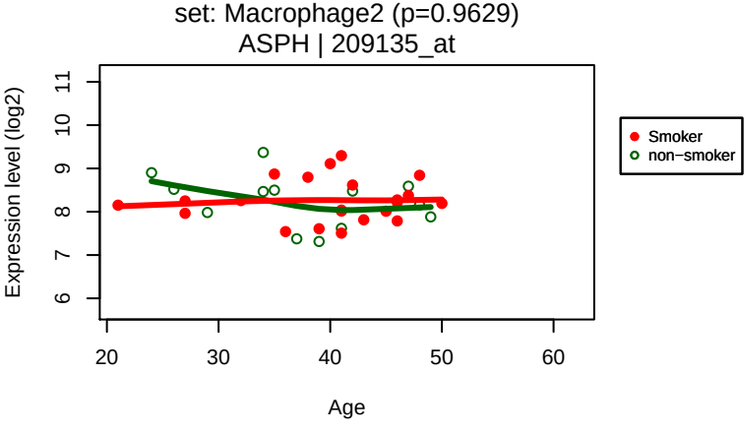

set: SmallAirway2 (p=7e−04)
ASPH | 209135_at
Expression level (log2)
Age
Smoker
non−smoker

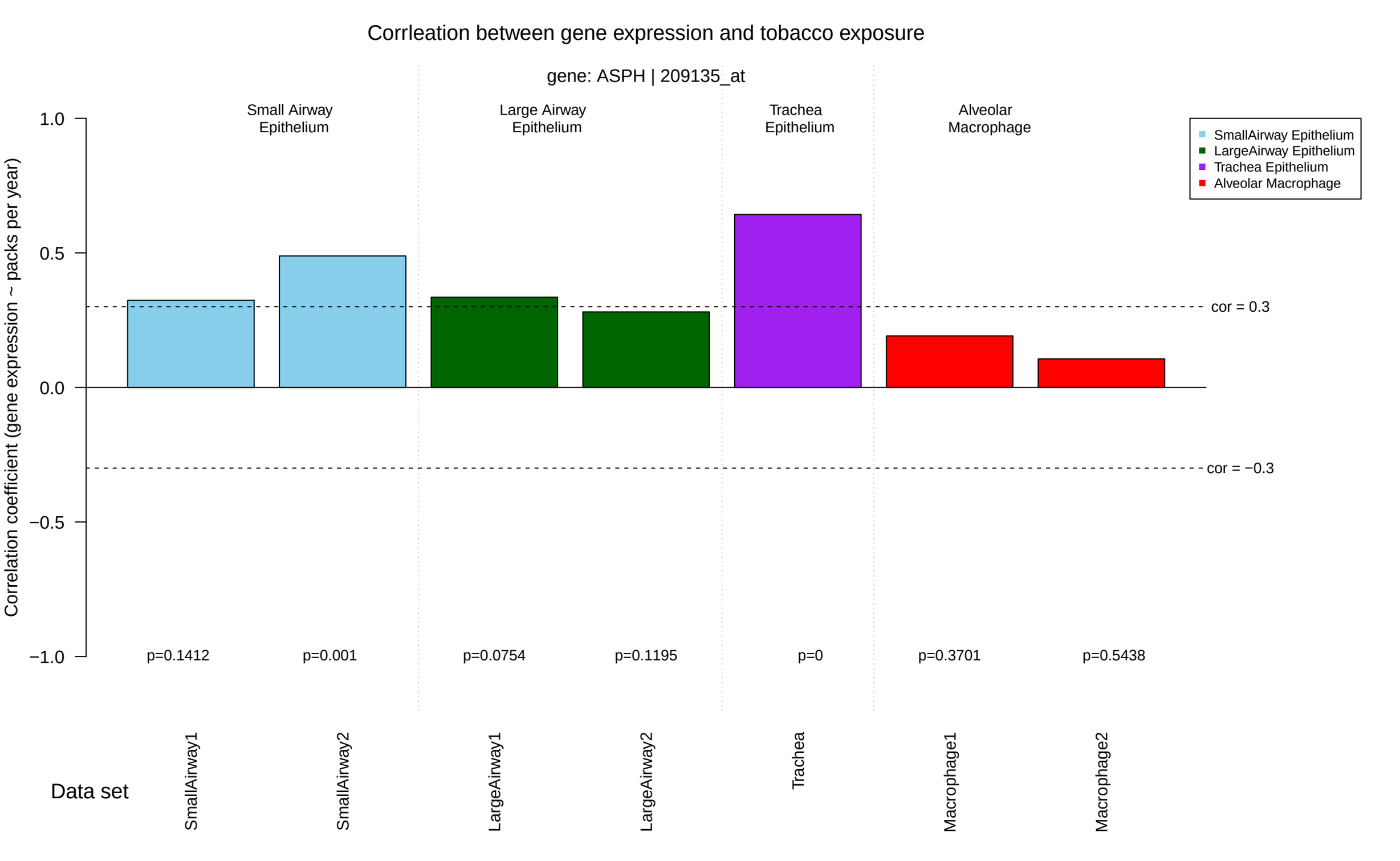

set: LargeAirway1 (p=0.0252)
ASPH | 209135_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2472)
ASPH | 209135_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
ASPH | 209135_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3955)
ASPH | 209135_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9629)
ASPH | 209135_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASPH | 209135_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1412
p=0.001
p=0.0754
p=0.1195
p=0
p=0.3701
p=0.5438
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2